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Assignment of Maths

Ques

$$\begin{aligned} 10x - 7y + 3z + 5u &= 6 \\ -6x + 8y - z - 4u &= 5 \\ 3x + y + 4z + 11u &= 2 \\ 5x - 9y - 2z + 4u &= 7 \end{aligned}$$

1. Augmented Matrix

$$\left[\begin{array}{cccc|c} 10 & -7 & 3 & 5 & 6 \\ -6 & 8 & -1 & -4 & 5 \\ 3 & 1 & 4 & 11 & 2 \\ 5 & -9 & -2 & 4 & 7 \end{array} \right]$$

2. Apply Row operation (Gauss-Jordan)
convert matrix to Reduced Row Echelon
form (RREF)

$$R_1 \rightarrow \frac{R_1}{10}$$

$$\left[\begin{array}{ccccc} 1 & -0.7 & 0.3 & 0.5 & 0.6 \\ -6 & 8 & -1 & -4 & 5 \\ 3 & 1 & 4 & 11 & 2 \\ 5 & -9 & -2 & 4 & 7 \end{array} \right]$$

$$\begin{aligned} R_2 &= R_2 + 6R_1 \\ R_3 &= R_3 - 3R_1 \\ R_4 &= R_4 - 5R_1 \end{aligned}$$

$$\left[\begin{array}{ccccc} 1 & -0.7 & 0.3 & 0.5 & 0.6 \\ 0 & 3.8 & 0.8 & -1 & 8.6 \\ 0 & 3.1 & 3.1 & 9.5 & 0.2 \\ 0 & -5.5 & 3.5 & 1.5 & 4 \end{array} \right]$$

Divide R_2 by 3.8

$$R_2 \rightarrow \frac{R_2}{3.8}$$

$$\begin{bmatrix} 1 & -0.7 & 0.3 & 0.5 & 0.6 \\ 0 & 1 & 0.21 & -0.21 & 2.26 \\ 0 & 3.1 & 3.1 & 9.5 & 0.2 \\ 0 & -5.5 & -3.5 & 1.5 & 4 \end{bmatrix}$$

$$R_1 = R_1 + 0.7 R_2$$

$$R_3 = R_3 - 3.1 R_2$$

$$R_4 = R_4 + 5.5 R_2$$

solve the follow

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -7 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$x = 5$$

$$y = 4$$

$$z = -7$$

$$u = 1$$

Ques 2 Solve using Gauss-Seidel Method

$$\begin{aligned} 1. \quad & 10x - 2y - z - u = 3 \\ & -2x + 10y - z - u = 15 \\ & -x - y + 10z + 2u = 27 \\ & -x - y - 2z + 10u = -9 \end{aligned}$$

convert into iterative form

$$x = \frac{3 + 2y + z + u}{10}$$

$$y = \frac{15 + 2x + z + u}{10}$$

$$z = \frac{27 + x + y - 2u}{10}$$

$$u = \frac{-9 + x + y + 2z}{10}$$

2. Initial values,

$$x_0 = 0$$

$$y_0 = 0$$

$$z_0 = 0$$

$$u_0 = 0$$

1.
$$x_1 = \frac{3 + 0 + 0 + 0}{10} = 0.3$$

$$y_1 = \frac{15 + 2(0.3) + 0 + 0}{10} = 1.56$$

$$z_1 = \frac{27 + 0.3 + 1.56 - 0}{10} = 2.886$$

$$u_1 = \frac{-9 + 0.3 + 1.56 + 2(2.886)}{10} = -0.1368$$

2.
$$x_2 = \frac{3 + 2(1.56) + 2.886 - 0.1368}{10} = 0.827$$

$$y_2 = \frac{15 + 2(0.827) + 2.886 - 0.1368}{10} = 1.95$$

$$x_1 = 1, y_2 = 2, z = 3, u = 0$$

$$3. \quad x_1 = \frac{2 + 2(0) + 0 + 0}{10}$$

$$= 0.3$$

$$y_1 = \frac{15 + 2(0.3) + 0 + 0}{10}$$

$$y_1 = \frac{15 + 0.6}{10}$$

$$y_1 = 1.56$$

$$z_1 = \frac{27 + 0.3 + 1.56 + 2(0)}{10}$$

$$z_1 = \frac{28.86}{10}$$

$$z_1 = 2.886$$

$$u_1 = \frac{-9 + 0.3 + 1.56 + 2(2.886)}{10}$$

$$= \frac{-9 + 0.3 + 1.56 + 5.772}{10}$$

$$u_1 = \frac{-1.368}{10}$$

$$u_1 = -0.1368$$

$$4. \quad x_2 = \frac{3 + 2(1.56) + 2.886 - 0.1368}{10}$$

$$= \frac{3 + 3.12 + 2.886 - 0.1368}{10} = \frac{8.8692}{10} = 0.8869$$

$$y_2 = \frac{15 + 2(0.8869) + 2.886 - 0.1318}{10}$$

$$= \frac{15 + 1.7738 + 2.886 - 0.1318}{10}$$

$$= \frac{19.528}{10}$$

$$y_2 = 1.9528$$

$$u_2 = \frac{-9 + 0.8869 + 1.9528 + 2(3 - 0.113)}{10}$$

$$u_2 = \frac{-9 + 0.8869 + 1.9528 + 6.0226}{10}$$

$$u_2 = \frac{-0.1382}{10}$$

$$= -0.0138$$

$$5. \quad x_3 = \frac{3 + 2(1.9528) + 3.0113 - 0.0138}{10}$$

$$= 0.9902$$

$$y_3 = \frac{15 + 2(0.9902) + 3.0113 - 0.0138}{10}$$

$$= 1.9978$$

$$z_3 = \frac{27 + 0.9902 + 1.9978 - 2(-0.0138)}{10}$$

$$= 3.0015$$

$$u_3 = \frac{-9 + 0.9902 + 1.9978 + 2(3.0015)}{10}$$

$$= -0.0011$$

Ques 32

Ques 32

Hug
dis

uu

Siri
vtr
if

ans

lag

$$x=1, y=2, z=3, u=0$$

Ques 32

x	y	z	w
0	0	0	0
0.3	1.56	2.886	-0.1368
0.8869	1.9523	3.0113	-0.0138
0.9902	1.9978	3.0015	-0.0011

$$x=1, y=2, z=3, u=0.$$

Ques 33 Interpolation problem -

Height x ft)	100	150	200	250	300	350	400
distance	10.63	13.63	15.64	16.81	18.42	19.90	21.7

we need to find

y when x=160

y when x=410

Since the values are not equally spaced around the req. points, we use Lagrange interpolation formula.

Ans Lagrange interpolation formula

$$y(x) = y_0 L_0 + y_1 L_1 + y_2 L_2$$

$$L_0 = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$$

$$L_y = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

(i) find y , when $x=160$

$$x_0=100, x_1=150, x_2=200$$

$$y_0=10.63, y_1=13.03, y_2=15.04$$

1. Calculate L_0

$$L_0 = \frac{(160-150)(160-200)}{(100-150)(100-200)}$$

$$L_0 = \frac{(10)(-40)}{(-50)(-100)}$$

$$L_0 = \frac{-400}{5000}$$

$$L_0 = -0.08$$

2.

L_1 ,

$$L_1 = \frac{(160-100)(160-200)}{(150-100)(150-200)}$$

$$= \frac{(60)(-40)}{(50)(-50)}$$

$$= \frac{-2400}{-2500}$$

$$= 0.96$$

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3. Cal, L_2

4. Substitue

(ii) find y

Take

(i) L_0, L_1

3. L_2

$$L_2 = \frac{(160-100)(160-150)}{(200-100)(200-150)}$$

$$L_2 = \frac{(60)(16)}{(100)(50)}$$

$$= \frac{600}{5000}$$

$$= 0.12$$

4. Substitute in formula

$$y = 10.63(-0.08) + 13.03(0.96) + 15.04(0.12)$$

$$y = -0.8504 + 12.5088 + 1.8048$$

$$y = 13.46$$

Result $y(160) \approx 13.46$

(ii) find y when $x = 410$

Take nearest values

$$x_0 = 300, x_1 = 350, x_2 = 400$$

$$y_0 = 18.42, y_1 = 19.90, y_2 = 21.27$$

(i) L_0

$$L_0 = \frac{(410-350)(410-400)}{(300-350)(300-400)}$$

$$L_0 = \frac{(60)(10)}{(-50)(-100)} = \frac{600}{5000} = 0.12$$

(2) d_1 cal.

$$L_1 = \frac{(410 - 300)(410 - 400)}{(350 - 300)(350 - 400)}$$
$$= \frac{(110)(10)}{(50)(-50)}$$

$$= \frac{1100}{-2500}$$

$$d_1 = -0.44$$

(3) d_2 cal

$$d_2 = \frac{(410 - 300)(410 - 350)}{(400 - 300)(400 - 350)}$$
$$= \frac{(110)(60)}{(100)(50)}$$

$$L_2 = \frac{6600}{5000}$$

$$L_2 = 1.32$$

(4)

$$y = 18.42(D.12) + 19.90(-0.44) + 21.22(1.32)$$
$$y = 2.2164 - 8.756 + 28.0764$$
$$y = 21.53$$

$$y(110) \approx 13.46$$

$$y(410) \approx 21.53$$

Ques Draw the DFA accepting the lang. over the alphabet $\{0,1\}$ that have the set of all strings that either begins or end (or both) with 01.

Ans $\Sigma = \{0,1\}$

The lang. consists of all string with 01 or end with 01.

state $q_0 \rightarrow$ Initial
 $q_1 \rightarrow$
 q_2
 q_3
 q_4

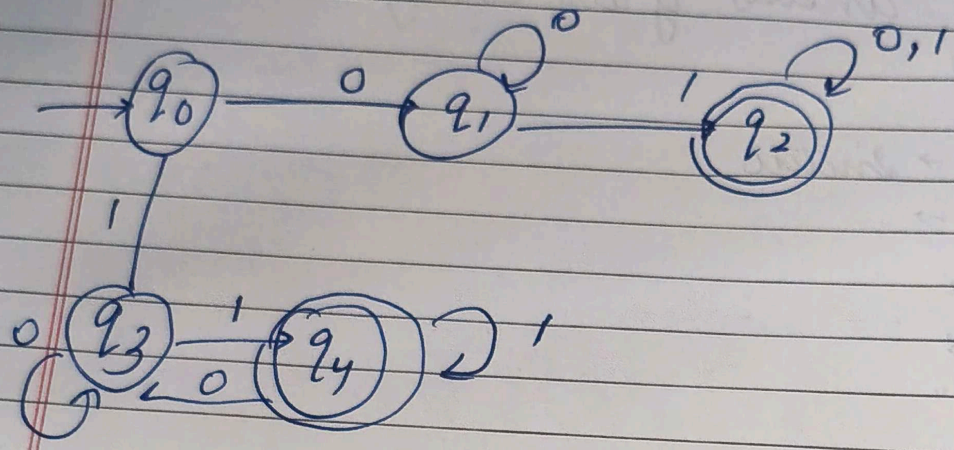
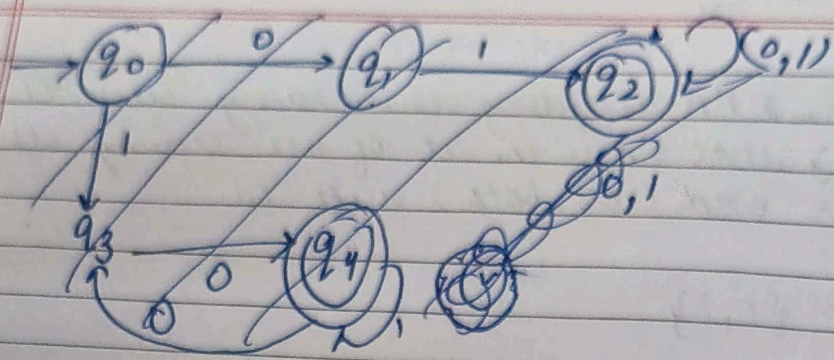
final states $F = \{q_2, q_4\}$

Transition Table.

q present	Input 0	Input 1
q_0	q_1	q_3
q_1	q_1	q_2
q_2	q_2	q_2
q_3	q_3	q_4
q_4	q_4	q_4

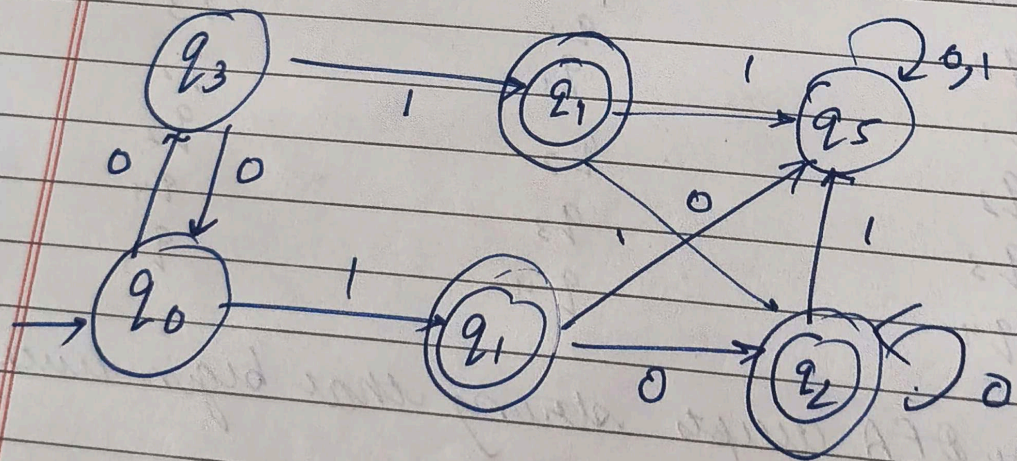
Thus, the DFA accepts string that begin with 01 or end with 01.

DFA



Ques 22 Minimize the follow DFA

Ans 12 final



final states

{q1, q2, q4}

Non-final states

$\{q_0, q_3, q_5\}$

2. Initial state.

$$P_1 = \{q_1, q_2, q_4\}, \{q_0, q_3, q_5\}$$

final equivalent states

$$q_1 = q_2$$

$$q_0 = q_3$$

4. Minimized state
new DFA states become.

$$A = \{q_0, q_3\}$$

$$B = \{q_1, q_2\}$$

$$C = \{q_4\}$$

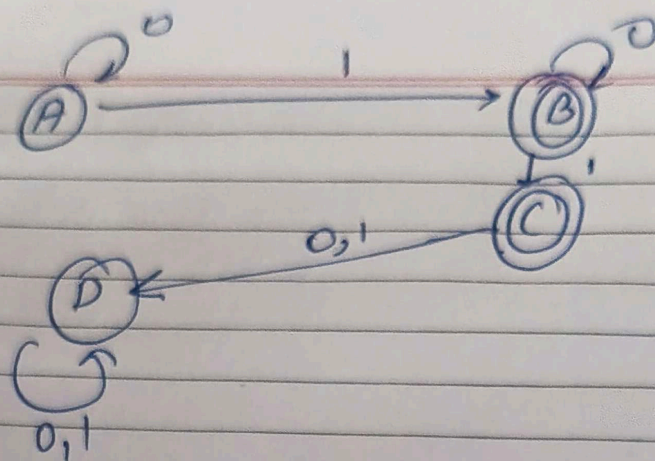
$$D = \{q_5\}$$

Minimized DFA

state	Input 0	1
A	A	B
B	B	C
C	D	D
D	D	D

start state = A

final $q = B, C$



Ques 3)

Convert Moore machine to Mealy machine

Ans

Moore machine output

state	output
q ₀	0
q ₁	0
q ₂	0
q ₃	1
q ₄	1

Conversion Rule,

Moore Machine, O/P is associated with states

Mealy machine, O/P is associated with transition

Mealy Machine transition table

present state	input	Next State	Out
q ₀	0	q ₁	0
q ₀	1	q ₂	0
q ₁	0	q ₁	1
q ₁	1	q ₃	1
q ₂	0	q ₄	0
q ₂	1	q ₂	0
q ₃	0	q ₁	1
q ₃	1	q ₁	0
q ₄	0	q ₂	1
q ₄	1	q ₃	

Ques 12 Explain Chomsky classification of lang. taking suitable eg of each classification?

Ans: Chomsky classification divides formal lang. into four types

1. Type 0 :- unrestricted grammar
Most general grammar
Recognized by Turing Machine

form

$$\alpha \rightarrow \beta$$

eg. $\Sigma = \{a^n b^n c^n \mid n \geq 1\}$

2. Type-1 - Context sensitive grammar

Rules depend on context

$$\forall A\beta \rightarrow \alpha\gamma\beta$$

Recognized by Linear Bounded Automata

$$\Sigma = \{a^n b^n c^n\}$$

2. Context free.

Rules $A \rightarrow \gamma$

Recognized by pushdown Automata

eg. $L = \{a^n b^n\}$
grammar

$S \rightarrow a S b / \epsilon$

3. Regular Grammar

simplest grammar

$A \rightarrow a b / a$

Recognized by finite Automata

eg. $L = \{a^* b^*\}$

Chomsky.

Regular $\subset$ Context free $\subset$ Context sensitive
 $\subset$ unrestricted.

Ans 3

Convert NFA to DFA.

Ans

states

$\{a, b, c, d, e\}$

start state

a

final state

e

NFA Transition Table.

state	0	1
a	a, c, b, d, e	d, e
b	c	e
c	a	b
d	e	-
e	-	-

subset construction method

{a}

DFA Transition Table

DFA	0	1
{a}	{a, b, c, d, e}	{d, e}
{a, b, c, d, e}	{a, b, c, d, e}	{b, d, e}
{d, e}	{e}	ϕ
{e}	ϕ	ϕ
ϕ	ϕ	ϕ

final states are any states containing e

DFA

